Hackathon Submission: Catch the Invisible AI Cheater

Track: Interview Integrity | Project: IntegrityGuard AI | Author: AK Srivastava

1. Project Overview & Architecture

IntegrityGuard AI is a zero-install browser-native prototype designed to detect candidates secretly using real-time AI interview copilots (such as Cluely, Parakeet, or screen teleprompters). Traditional proctoring software is easily bypassed by these tools because they run as transparent system overlays or on separate devices. Our system operates entirely within the interview tab on localhost, simulating a real-time mock AI interviewer using browser speech APIs while passively streaming behavioral telemetry to a separate reviewer console.

2. Core Detection & Telemetry Channels

Rather than relying on invasive screen recording or system scans, we monitor candidate behavior through four core channels:

- **Speech Latency (Timing)**: Measures the exact time gap between the end of the question voice playback and the candidate's first spoken word. Cheating tools create an inevitable 5-8+ second delay to generate answers, whereas natural speech begins within 1.5-2.5 seconds.
- **Calibrated Iris Deviation (Gaze)**: Uses client-side MediaPipe FaceMesh to track iris coordinate movement relative to the eye socket corners. On interview start, the system runs a 50-frame baseline calibration. This allows natural monitor-reading shifts, while capturing large vertical or horizontal deflections (like reading from a phone or second display).
- **Tab Focus Tracking (Blurs)**: Registers events when the candidate switches tabs or clicks outside the interview window (e.g. to copy/paste text or search coordinates).
- **AI-Likeness Phrasing Scorer**: Evaluates verbal transcript structure (via Grok API or offline keyword heuristics) for AI-generated written styles (e.g., transitional lists, lack of speech fillers). Short transcripts (<15 words) are bypassed to prevent false flags.

3. Multi-Modal Risk Fusion & Reviewer Experience

To satisfy hackathon requirements, the system **never auto-rejects** candidates. Telemetry is fused by a risk aggregator into a composite score and classification (Clean, Suspicious, or High Risk). High-severity overrides are implemented to trigger alerts immediately if a single signal is definitive (e.g., Pause > 10s, Gaze > 50%, or Tab Blur >= 1). The reviewer dashboard renders raw telemetry numbers, transcript snippets, and AI rationale logs in real time, providing a transparent audit trail.

4. Honest Coverage & System Limitations

IntegrityGuard AI acts as a smart behavioral signal aggregator, not an absolute verdict tool. Understanding its limits is crucial:

- **What it Catches**: Real-time LLM query-generation delays; constant reading shifts from screen overlays or second monitors; physical glances down to check a phone; tab departure focus losses.
- **System Boundaries**: It cannot detect an off-screen collaborator whispering answers if the candidate is articulate; pre-memorized answers; or candidates with strong physical reading discipline. Low room lighting or backlighting can also degrade FaceMesh tracking accuracy.
- **Citations & Open Source**: Built using MediaPipe FaceMesh WASM client, webkitSpeechRecognition API, FastAPI WebSockets, and ReportLab. It contains zero third-party agent installs or extensions.